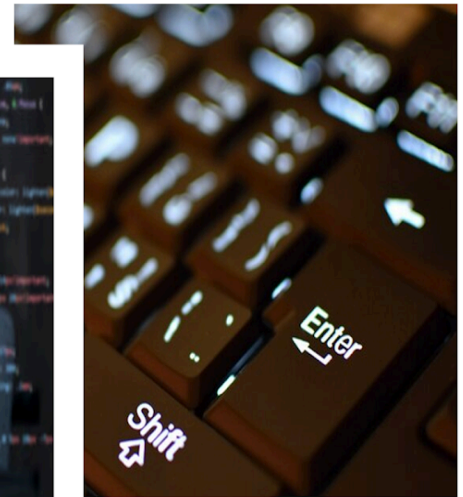


Computer Awareness

STUDY NOTES

Computer Fundamental



Computer Fundamentals

Functionalities of Computer

If we consider it in a very broad sense, any digital computer performs the following five operations –

- Step 1 – Accepts data as input.
- Step 2 – Saves the data/instructions in its memory and utilizes them as required.
- Step 3 – Execute the data and convert it into useful information.
- Step 4 – Provides the output.
- Step 5 – Have control over all the above four steps.

Advantages of Computer

- **High Speed**
 - ◇ Computers are very fast devices.
 - ◇ It can perform calculations of a very huge amount of data.
 - ◇ The computer has units of speed in microsecond, nanosecond, and even the pico second.
 - ◇ It can process millions of computational calculations in a fraction of seconds unlike a man who may spend many months performing the same task.
- **Accuracy**
 - ◇ In addition to being very fast, computers are very accurate.
 - ◇ The calculations are 100% error free.
 - ◇ Computers perform all jobs with 100% accuracy provided that the input is correct.
- **Storage Capability**
 - ◇ Memory is a very important characteristic of computers.
 - ◇ A computer has much more storage capacity than human beings.
 - ◇ It can store large amounts of data.
 - ◇ It can store any type of data such as images, videos, text, audio, etc.
- **Diligence**
 - ◇ Unlike human beings, a computer is free from monotony, tiredness, and lack of concentration.
 - ◇ It can work continuously without any error and boredom.
 - ◇ It can perform repeated tasks with the same speed and accuracy.

- **Versatility**

- ◇ A computer is a very versatile machine.
- ◇ A computer is very flexible in performing the jobs to be done.
- ◇ This machine can be used to solve problems related to various fields.
- ◇ At one moment, it may be processing and simplifying a complex scientific problem and the very next moment it may be running a card game.

- **Reliability**

- ◇ A computer is a reliable machine.
- ◇ Modern electronic components have long lives.
- ◇ Computers are designed to make maintenance easy.

- **Automation**

- ◇ A computer is an automatic machine.
- ◇ Automation is the capability due to which a machine can perform its task without need of human consideration. Once the program is fed to the computer i.e., the program is held in the computer memory, then the program and instruction can transfer the program execution in absence of human consideration.

- **Reduction in Paper Work and Cost**

- ◇ With the help of computers for data processing in an organization, there is reduction of paper work and results in speeding up the process, which in return saves trees.
- ◇ Because data saved as electronic files can be retrieved whenever required, the headache of maintaining a pile number of paper files gets reduced.
- ◇ While the investment at the beginning for installation of a computer is high, it substantially reduces the cost of each of its transactions.

Disadvantages of Computer

- **No I.Q.**

- ◇ A computer is a machine that has no intelligence to perform any task.
- ◇ Each instruction has to be given to the computer.
- ◇ A computer is not capable enough to make any decision on its own.

- **Dependency**

- ◇ It works according to the user's instruction, thus it is entirely dependent on humans.

- **Environment**

- ◇ The operating environment of the computer should be dust free and suitable.

- **No Feeling**

- ◇ Computers have no feelings or emotions.
- ◇ It is not smart enough to make judgment based on experience, feeling, taste, and knowledge just like humans.

Applications

Business

- Because a computer system has a very high calculation speed, diligence, accuracy, reliability, or versatility, it became an integrated part in all business institutions.
- Computer is very helpful in business institutions in terms of –
 1. Payroll calculations
 2. Budgeting
 3. Sales analysis
 4. Financial forecasting
 5. Managing employee database
 6. Maintenance of stocks, etc.

Banking

- In the present time, banking is entirely based on computers.
- Banks provide the following facilities –
 - ◇ Online accounting facility, which includes checking current balance, making deposits and overdrafts, checking interest charges, shares, and trustee records.
 - ◇ ATM machines which are entirely automated are making this facility simpler for customers to deal with banks.

Insurance

- Insurance companies are handling all records and keeping them up-to-date using computers. Entities like Insurance companies, finance houses, and stock broking firms, etc., are using computers at a very high scale for their concerns.
- Companies are keeping a database of all customers with information showing –
 1. Procedure to continue with policies

2. Starting date of the policies
3. Next due instalment of a policy
4. Maturity date
5. Interests due
6. Survival benefits
7. Bonus

Education

- The computer helps in providing a lot of facilities in the education system.
- There is a tool available on computers for the education system known as CBE (Computer Based Education).
- CBE has many features that can be used for the purpose of control, delivery, and evaluation of learning.
- Education based on computers is transforming the way students study and in turn increasing the graph of the number of computer students.
- A lot of ways in which educational institutions can utilize a computer to educate the students.
- It helps to create a centralized database which stores the performance of a student and analysis is carried out on this basis.

Marketing

- **In advertising** – Using computers, **professionals working in advertising** make **art and graphics, write and revise copy, and print and disseminate ads** with the purpose of selling more products.
- **Home Shopping** – It is possible to shop while sitting at home through the use of computerized catalogues that facilitate access to product information and allow direct entry of orders to be filled by the customers.

Healthcare

- The use of computers has become an integral part at hospitals, labs, and dispensaries.
- They are being used in hospitals to keep the record of patients and medicines.
- It also aids in scanning and diagnosing various diseases. ECG, EEG, ultrasounds and CT scans, etc. are also done by computerized machines.
- Computers are used in following major fields of health care:
 1. **Diagnostic System** – Computers are used to collect data and identify

the cause of illness.

2. **Lab-diagnostic System** – All tests can be done and the reports are prepared by computer.
3. **Patient Monitoring System** – These are used to check the patient's signs for abnormality such as in Cardiac Arrest, ECG, etc.
4. **Pharma Information System** – Computer is used to check drug labels, expiry dates, harmful side effects, etc.
5. **Surgery** – In these days, computers are also used in conducting surgery.

Engineering Design

- Computers are widely used for engineering purposes.
- CAD (Computer Aided Design) is one of the major areas that aids in creation and modification of images. Some of the fields are-
 1. **Structural Engineering** – Needs digital stress and strain recreation and analysis for designing of ships, budgets, airplanes, buildings, etc.
 2. **Industrial Engineering** – Computers deal with design, implementation, and improvement of integrated systems of people, materials, and equipment.
 3. **Architectural Engineering** – Systems based on computers aid in planning towns, designing buildings, determination of a range of buildings on the sites with the help of both 2D and 3D drawings.

Military

Computers are extensively used in defence in designing and controlling modern tanks, missiles, weapons, etc. They also deploy computerized control systems that are used for communication. Some areas in military where a computer has been used are –

1. Missile Control
2. Military Communication
3. Military Operation and Planning
4. Smart Weapons

Communication

It is a process to convey a message, an idea, a picture, or speech that is received and understood clearly and correctly by the person for whom it is meant. Some main areas in this category are –

1. E-mail
2. Chatting

3. Usenet
4. FTP
5. Telnet
6. Video-conferencing

Government

Computers perform a very important role in services related to the government. Some major fields in this category are –

1. Budgets
2. Sales tax department
3. Income tax department
4. Computation of male/female ratio
5. Computerization of voters lists
6. Computerization of PAN card
7. Weather forecasting

Generations of Computer

- The generations in computer terminology describes a change in technology (on which the computer system is designed) a computer is/was being used. Basically, the generation word was used to differentiate between various hardware technologies. Nowadays, generation encapsulates both hardware and software, which in combination make up an entire computer system.
- There have been 5 computer generations till date. In the following list, approximate dates against each generation have been mentioned, which are normally accepted.
- There are five main generations of computers based on their processing hardware.
 1. First Generation: period: 1946-1959. Vacuum tube based.
 2. Second Generation: period: 1959-1965; Transistor based.
 3. Third Generation: period: 1965-1971; Integrated Circuit based.
 4. Fourth Generation: period: 1971-1980; VLSI microprocessor based.
 5. Fifth Generation: period: 1980-onwards; ULSI microprocessor based.